

7 Factor Analysis: Theoretical Models

MA2003B Application of Multivariate Methods
in Data Science

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The goal is to reduce complexity by grouping related variables into a smaller number of underlying **factors**, which are easier to interpret.

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- **Finance and economics:** to reveal hidden drivers of stock returns, macroeconomic indicators, or risk factors.
- **Education:** to analyze test items and determine whether they measure common skills or abilities.
- **Biological and health sciences:** to find latent structures in genetic data, symptom checklists, or patient-reported outcomes.

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It is especially useful for building measurement scales, data reduction, and gaining insights into the structure of complex datasets.

Factor Analysis Setting

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$$X = \mu + LF + \varepsilon$$

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- $\varepsilon \sim \mathcal{N}(0, \Psi)$ is a vector of unique errors (specific variances), with Ψ diagonal.
- F and ε are assumed to be independent, in particular $\text{Cov}[F, \varepsilon] = 0$.

As a consequence of these assumptions, we have that

$$\mathbb{E}[X] = \mathbb{E}[\mu + LF + \varepsilon] = \mathbb{E}[\mu] + L\mathbb{E}[F] + \mathbb{E}[\varepsilon] = \mu$$

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and

$$\begin{aligned}\text{Var}[X] &= \text{Var}[LF + \varepsilon] \\ &= \mathbb{E}[(LF + \varepsilon)(LF + \varepsilon)^T] \\ &= \mathbb{E}[LFF^T L^T] + \mathbb{E}[\varepsilon\varepsilon^T] + \mathbb{E}[LF\varepsilon^T] + \mathbb{E}[\varepsilon F^T L^T] \\ &= L\mathbb{E}[FF^T]L^T + \mathbb{E}[\varepsilon\varepsilon^T] + L\mathbb{E}[F\varepsilon^T] + \mathbb{E}[\varepsilon F^T]L^T \\ &= LI_m L^T + \Psi + L0 + 0L^T \\ &= LL^T + \Psi\end{aligned}$$

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Furthermore, since X is a linear combination of normally distributed variables, it follows that

$$X \sim \mathcal{N}(\mu, LL^T + \Psi)$$

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In particular, the variance of each individual observed variable X_i can be expressed as

$$\text{Var}[X_i] = \sum_{j=1}^m L_{ij}^2 + \sigma_i^2$$

where $h_i^2 = \sum_{j=1}^m L_{ij}^2$ is called the **communality** of the variable X_i , representing the portion of its variance explained by the common factors, and σ_i^2 is the unique variance specific to X_i .

Factor Score Estimation

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Then the joint distribution of X and F is given by

$$\begin{bmatrix} X \\ F \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu \\ 0 \end{bmatrix}, \begin{bmatrix} LL^T + \Psi & L \\ L^T & I_m \end{bmatrix} \right)$$

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Notice that, in particular,

$$\text{Cov}[X, F] = \mathbb{E}[(X - \mu)(F - 0)^T] = \mathbb{E}[(LF + \varepsilon)F^T] = L\mathbb{E}[FF^T] + \mathbb{E}[\varepsilon F^T] = LI_m + 0 = L$$

Using the properties of the multivariate normal distribution, we can derive the conditional distribution of the factors F given the observed variables X .

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This conditional distribution is also normal, with mean and covariance given by

$$[F|X = x] \sim \mathcal{N} \left(L^T (LL^T + \Psi)^{-1} (x - \mu), I_m - L^T (LL^T + \Psi)^{-1} L \right)$$

The mean of this conditional distribution,

$$\mathbb{E}[F|X = x] = L^T(LL^T + \Psi)^{-1}(x - \mu)$$

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The covariance matrix

$$I_m - L^T (LL^T + \Psi)^{-1} L$$

represents the uncertainty associated with the factor score estimates.

Remark

Notice that, if

$$f = \mathbb{E}[F|X = x] = L^T(LL^T + \Psi)^{-1}(x - \mu)$$

then, in general, the **residual**

$$r(x) = x - \mu - Lf$$

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Remark

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then, in general, the **residual**

$$r(x) = x - \mu - Lf$$

is different from zero.

In fact, as a function of the random variable X , the residual has probability distribution

$$r(X) \sim \mathcal{N}\left(0, L(I_m + L^T\Psi^{-1}L)^{-1}L^T + \Psi\right)$$

Rotation of Factors

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In particular,

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The corresponding factor loadings will be transformed as

$$L^* = LQ^T$$

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is a valid factor model, then so is

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In the next presentation, we will explore various practical methods for factor analysis and rotation techniques geared towards achieving more interpretable factor solutions.